

IN5340/IN9340 Project IV: Spectral Estimation

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Code: `github.com/Twaal/Statistical\_Signal\_Processing\_IN5340/tree/master/Project\_4`

Agenda

- 1 Introduction
- 2 Exercise 1A - Spectral Estimation (Channel 14)
- 3 Exercise 1B - Spectral Estimation (Channel 9)
- 4 Exercise 2A - Spectrogram Analysis (Channel 14)
- 5 Exercise 2B - Spectrogram Analysis (Channel 32)

Problem statement

- Sonar data from HUGIN AUV: $N_h = 32$ hydrophones, single ping.
- Data matrix $\mathbf{X} \in \mathbb{C}^{N_t \times 32}$, sampling frequency $f_s = 40$ kHz.
- Transmitted signal: LFM pulse, bandwidth $B = 30$ kHz, centre frequency $f_c = 100$ kHz.

$$s_{Tx}(t) = \exp(j2\pi\alpha t^2/2), \quad |t| \leq T_p/2, \quad \alpha = B/T_p$$

- Received data are basebanded (carrier removed).
- Goal: estimate and compare power spectral densities using periodogram, modified periodogram, Welch, and multitaper methods; analyse spectrograms.

1A: Periodogram and modified periodogram

```
# periodogram
fft_result = np.fft.fft(channel_14_data)
periodogram = (1/(N*fs)) * np.abs(fft_result)**2
periodogram_shifted = np.fft.fftshift(periodogram)

# modified periodogram with Hanning window
win = np.hanning(N)
U = np.mean(win**2)
windowed = channel_14_data * win
fft_win = np.fft.fft(windowed)
periodogram_windowed = (1/(N*fs*U)) * np.abs(fft_win)**2
```

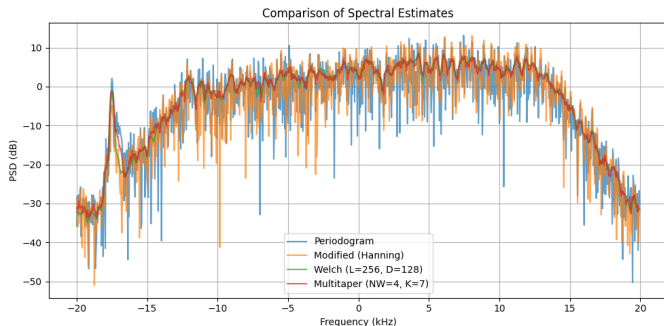
1A: Welch method and multitaper

```
# Welch (L=256, D=128, Hanning)
welch_psd = np.zeros(L)
for i in range(num_segments):
    start = i*(L - D)
    seg = channel_14_data[start:start+L] * welch_win
    welch_psd += (1/(L*fs*U_welch)) * np.abs(np.fft.fft(seg))**2
welch_psd /= num_segments

# Multitaper (NW=4, K=7, DPSS tapers)
tapers, eigenvalues = windows.dpss(N, NW, K, return_ratios=True)
multitaper_psd = np.zeros(N)
for k in range(K):
    multitaper_psd += (1/fs) * np.abs(np.fft.fft(channel_14_data * tapers[k]))**2
multitaper_psd /= K
```

1A: Comparison of spectral estimates

- Channel 14, $M = 1200$, $N = 1024$.
- Four estimators plotted: periodogram, modified (Hanning), Welch ($L = 256$, $D = 128$), multitaper ($NW = 4$, $K = 7$).
- The LFM passband (± 15 kHz) is visible in all estimates.
- Periodogram shows the most variance; Welch and multitaper are smoother.



1A: Bias and variance analysis

Does bias reduction help?

The boxcar periodogram is theoretically unbiased only for infinite data. With finite $N = 1024$, spectral leakage introduces bias. All methods show negative mean bias in the passband:

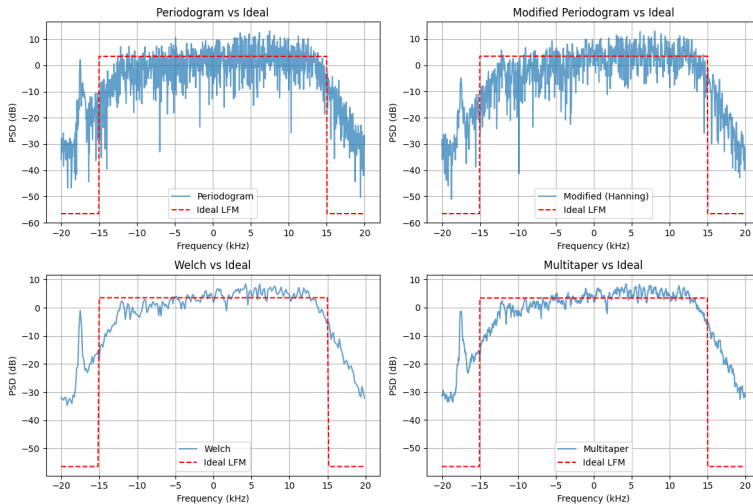
Method	Mean Bias (dB)
Periodogram	-3.86
Modified (Hanning)	-3.35
Welch ($L=256$)	-1.57
Multitaper ($K=7$)	-1.43

Does variance reduction help?

Method	Mean Variance
Periodogram	9.19
Modified (Hanning)	11.17
Welch ($L=256$)	0.81
Multitaper ($K=7$)	0.75

Welch and multitaper reduce variance by $> 10\times$, giving a much more reliable estimate.

1A: Estimators vs ideal LFM spectrum



1A: Bandwidth estimation and spectrum issues

Bandwidth estimation (multitaper, -6 dB from passband level):

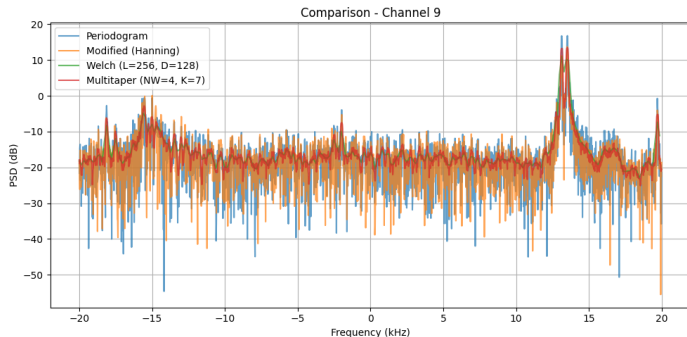
- Passband level: 3.45 dB
- -6 dB threshold: -2.55 dB
- Estimated edges: -17.58 kHz to 14.53 kHz
- **Estimated BW:** 32.11 kHz (ideal: 30 kHz)

What is wrong with the spectrum?

- Spectral leakage widens the apparent passband beyond the true 30 kHz.
- An interfering tone near -17 kHz occupies the lower band and skews the bandwidth estimate upward.

1B: Comparison of spectral estimates

- Channel 9, $M = 7000$, $N = 2048$.
- Signal dominated by additive noise and narrowband interference from "tonal lines".
- Periodogram resolves closely spaced tones near 16-17 kHz that merge in Welch/multitaper.



1. Does bias reduction help?

No. The signal is dominated by narrowband tones in noise, not a wideband LFM pulse. There is no well-defined passband where spectral leakage matters, so bias reduction provides no practical benefit.

2. Does variance reduction help?

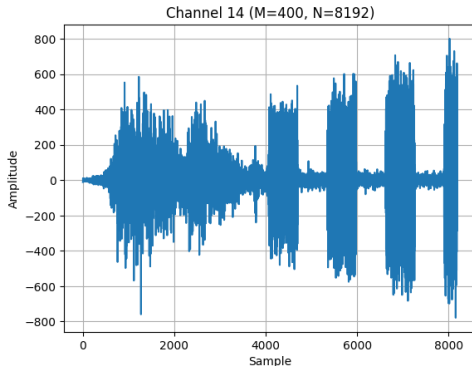
Variance reduction is counterproductive here. At 16-17 kHz two closely spaced tones are distinguishable in the high-variance periodogram but merge in the smoothed Welch estimate.

3. Consequence of variance reduction?

A bias-variance-resolution tradeoff: smoothing reduces variance but also reduces frequency resolution. The Welch method with $L = 256$ gives $\Delta f = f_s/L \approx 156$ Hz, which is too coarse to separate nearby tones.

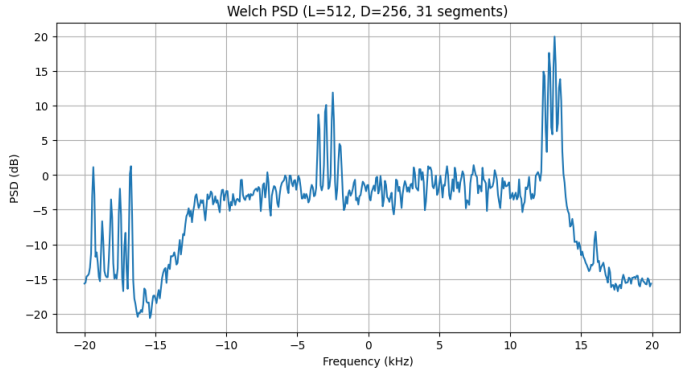
2A: Timeseries - Channel 14

- Channel 14, $M = 400$, $N = 8192$.
- Several distinct events with varying amplitude and duration in additive noise.
- Not WSS over the full window motivates time-frequency analysis.



2A: Welch PSD - Channel 14

- Welch with $L = 512$, $D = 256$, Hanning.
- Several tonal clusters visible.
- Original LFM passband present but tones dominate.
- The global PSD hides the time-varying nature of these events.



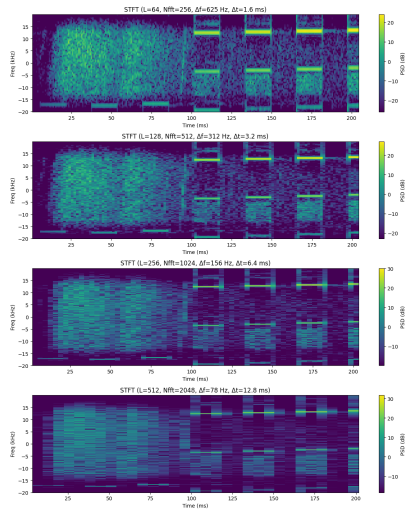
2A: STFT implementation

```
for idx, Ls in enumerate(seg_lengths):      # L = 64, 128, 256, 512
    Ds = Ls // 2
    Nfft = Ls * zp                          # zero-pad x4
    w = np.hanning(Ls); Uw = np.mean(w**2)

    nseg = (N_2a - Ls) // Ds + 1
    stft = np.zeros((Nfft, nseg))
    for i in range(nseg):
        s = i * Ds
        seg = ch14_long[s:s+Ls] * w
        padded = np.zeros(Nfft, dtype=complex)
        padded[:Ls] = seg
        stft[:, i] = (1/(Ls*fs*Uw)) * np.abs(np.fft.fft(padded))**2

    stft_dB = 10*np.log10(np.fft.fftshift(stft, axes=0) + 1e-30)
```

2A: STFT - four segment lengths



1. Consequence of segment length?

Segment length controls the time-frequency resolution tradeoff:

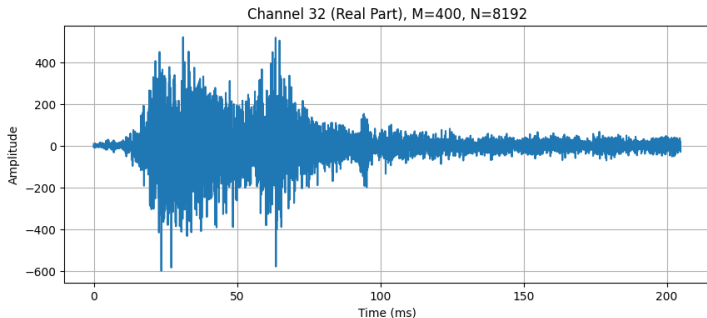
- Short L (64): good time resolution ($\Delta t = 1.6$ ms), poor frequency resolution ($\Delta f = 625$ Hz).
- Long L (512): good frequency resolution ($\Delta f = 78$ Hz), poor time resolution ($\Delta t = 12.8$ ms).

2. Interfering chips (100-200 ms)?

- The chips are approximately 20 ms in duration.
- Their centre frequencies show a slight shift from chip to chip, possibly due to Doppler from a radially accelerating interferer.

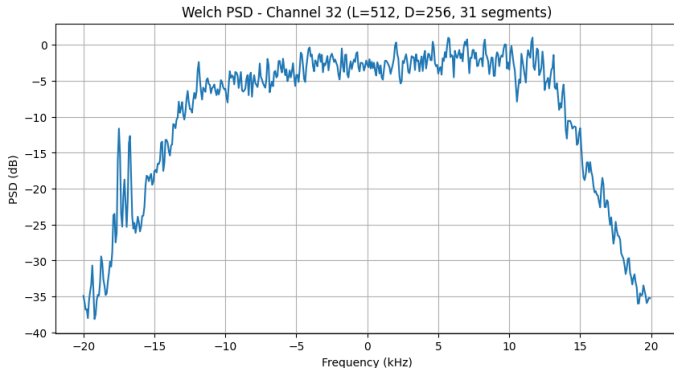
2B: Timeseries - Channel 32

- Channel 32, $M = 400$, $N = 8192$.
- Real part of the signal plotted.
- A burst of energy around 20 ms followed by decaying amplitude.

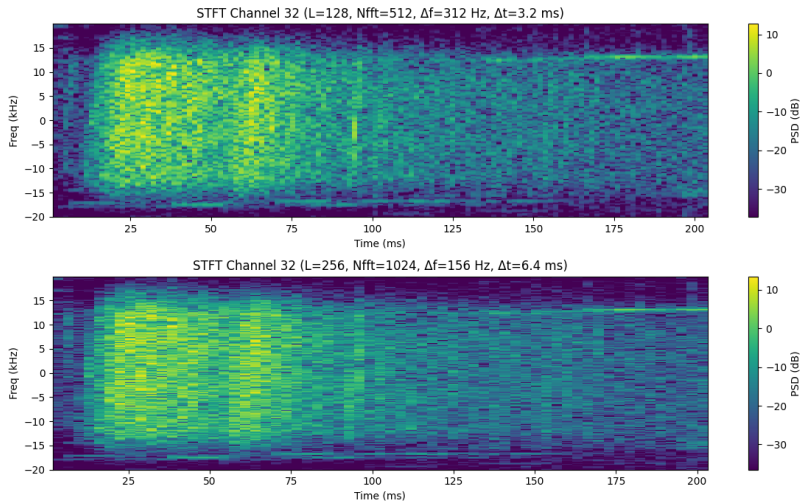


2B: Welch PSD - Channel 32

- Welch with $L = 512$, $D = 256$, Hanning.
- Spectrum resembles the LFM passband (± 15 kHz) with some leakage.
- Weak interfering tone near -17 kHz.
- Much cleaner than Channel 14.



2B: STFT - Channel 32



2B: Discussion - Channel 14 vs 32

Signal characteristics (STFT):

- The LFM echo is spread equally across ± 15 kHz, starting around 20 ms and fading slowly.
- A weak interfering tone near 15 kHz appears around 130 ms.
- The -17 kHz interferer from Channel 14 is barely visible here.

Key difference between channels:

- Channel 32 has much less interference from narrowband tones.
- The interferers that dominate Channel 14 are nearly absent in Channel 32.
- This spatial interference originates possibly from faulty electronics or near-field source rather than a far-field acoustic source which would appear similarly across all channels.

- **Ex 1A:** Welch and multitaper reduce variance by $> 10\times$ vs the periodogram, with lower passband bias. Estimated bandwidth is 32.1 kHz vss. ideal 30 kHz; spectral leakage and a -17 kHz interferer bias the estimate.
- **Ex 1B:** Variance reduction merges closely spaced tones. High resolution periodogram or windowed periodogram is preferred to resolve narrowband signals.
- **Ex 2A:** STFT reveals time-varying events hidden by global PSD. Shorter segments improve time resolution at the cost of frequency resolution.
- **Ex 2B:** Channel 32 shows a cleaner LFM echo with minimal interference, suggesting the tones in Channel 14 originate from near-field, likely not far-field sources.